# Chapter 10. Models for Matched Pairs

*McNemar, symmetry, agreement, Bradley-Terry*

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Matched pairs (repeated measurement, case–control matching, two raters) give a square $I \times I$ table of dependent responses.

## 10.1 Comparing dependent proportions

**Motivation.** Matched proportions are correlated; the correct test of equal margins uses only the discordant pairs.

For $n$ pairs let $\pi_{ab}$ be the probability of outcome $a$ for observation 1 and $b$ for observation 2, $n_{ab}$ the count, $p_{ab} = n_{ab}/n$, and $\{n_{ab}\} \sim \text{multinomial}(n; \{\pi_{ab}\})$.

**Definition (Marginal homogeneity)**

The square table satisfies **marginal homogeneity** if $\pi_{a+} = \pi_{+a}$ for all $a$.

For $2 \times 2$ tables $\pi_{1+} - \pi_{+1} = \pi_{12} - \pi_{21}$, so marginal homogeneity means $\pi_{12} = \pi_{21}$.

### 10.1.1 Inference for dependent proportions

Let $\delta = \pi_{+1} - \pi_{1+}$ and $d = p_{+1} - p_{1+} = p_{21} - p_{12}$.

**Theorem 10.1 (Variance of a difference of dependent proportions)**

Under multinomial sampling,

$$\begin{aligned} n \operatorname{Var}(d) &= \pi_{1+}(1-\pi_{1+}) + \pi_{+1}(1-\pi_{+1}) - 2(\pi_{11}\pi_{22} - \pi_{12}\pi_{21}) \\ &= (\pi_{12} + \pi_{21}) - (\pi_{12} - \pi_{21})^2. \end{aligned}$$

Hence $d \pm z_{\alpha/2}\,\hat{\sigma}(d)$ with $\hat{\sigma}^2(d) = \left[(p_{12} + p_{21}) - (p_{12} - p_{21})^2\right]/n$ is a large-sample confidence interval for $\delta$.

**Proof.**

*Step 1.* Multinomial moments: $n \operatorname{Var}(p_{ab}) = \pi_{ab}(1 - \pi_{ab})$, $n \operatorname{Cov}(p_{ab}, p_{cd}) = -\pi_{ab}\pi_{cd}$ for distinct cells. Each margin is binomial, and

$$n \operatorname{Cov}(p_{11} + p_{21},\, p_{11} + p_{12}) = \pi_{11}(1 - \pi_{11} - \pi_{12} - \pi_{21}) - \pi_{12}\pi_{21} = \pi_{11}\pi_{22} - \pi_{12}\pi_{21},$$

so $\operatorname{Var}(p_{+1}) + \operatorname{Var}(p_{1+}) - 2\operatorname{Cov}(p_{+1}, p_{1+})$ gives the first line.

*Step 2.* Directly, $n \operatorname{Var}(p_{21} - p_{12}) = \pi_{21}(1 - \pi_{21}) + \pi_{12}(1 - \pi_{12}) + 2\pi_{12}\pi_{21}$, which is the second line.

*Step 3.* By the multivariate CLT and consistency of $\hat{\sigma}(d)$, Slutsky's theorem gives the interval. ■

Under $H_0 : \delta = 0$ the variance becomes $(\pi_{12} + \pi_{21})/n$, estimated by $\hat{\sigma}_0^2(d) = (n_{12} + n_{21})/n^2$.

**Theorem 10.2 (McNemar's test is the score test)**

The score statistic for $H_0 : \pi_{12} = \pi_{21}$ in the multinomial model is $z_0^2$, where

$$z_0 = \frac{d}{\hat{\sigma}_0(d)} = \frac{n_{21} - n_{12}}{(n_{21} + n_{12})^{1/2}}; \tag{10.1}$$

$z_0^2$ is asymptotically $\chi_1^2$ under $H_0$ (**McNemar's test**).

**Proof.**

*Step 1.* Reparametrize by $(\pi_{11}, \pi_{22}, \sigma, q)$ with $\sigma = \pi_{12} + \pi_{21}$, $q = \pi_{21}/\sigma$, subject only to $\pi_{11} + \pi_{22} + \sigma = 1$; $H_0$ is $q = \frac{1}{2}$. With $n^* = n_{12} + n_{21}$,

$$L = n_{11} \log \pi_{11} + n_{22} \log \pi_{22} + n^* \log \sigma + n_{21} \log q + n_{12} \log(1 - q),$$

and $q$ separates from the constrained parameters, so the information is block diagonal and the score test uses only the $q$-block.

*Step 2.* $U(q) = n_{21}/q - n_{12}/(1-q)$ and $\mathcal{I}(q) = n\sigma/[q(1-q)]$. At $q = \frac{1}{2}$, $\hat{\sigma} = n^*/n$: $U = 2(n_{21} - n_{12})$, $\mathcal{I} = 4n^*$, and $U^2/\mathcal{I} = (n_{21} - n_{12})^2/n^* = z_0^2$. Also $d/\hat{\sigma}_0(d) = [(n_{21} - n_{12})/n]/[(n^*)^{1/2}/n]$ gives (10.1); the $\chi^2_1$ limit is standard score asymptotics. ■

The diagonal counts do not affect the test, though all pairs enter $d$ and its standard error.

### 10.1.2 Prime Minister approval rating example

**Example 10.3 (Prime Minister approval)**

1600 British voters rated the Prime Minister twice, six months apart: $(n_{11}, n_{12}, n_{21}, n_{22}) = (794, 150, 86, 570)$ (1 = approve). $p_{1+} = 0.59$, $p_{+1} = 0.55$, $\hat{\sigma}(d) = 0.0095$, 95% interval for $\pi_{+1} - \pi_{1+}$: $(-0.06, -0.02)$. McNemar $z_0 = (86 - 150)/236^{1/2} = -4.17$: approval dropped.

### 10.1.3 Increased precision with dependent samples

Independent samples have $n \operatorname{Var}(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1(1 - \pi_1) + \pi_2(1 - \pi_2)$ (no covariance term). Matched data are usually positively dependent ($\pi_{11}\pi_{22} > \pi_{12}\pi_{21}$), reducing $\operatorname{Var}(d)$: SE 0.0095 versus 0.0175 in Example 10.3.

### 10.1.4 Small-sample test comparing matched proportions

**Proposition 10.4 (Exact conditional test)**

Conditional on $n^* = n_{12} + n_{21}$, $n_{21} \sim \text{bin}(n^*, \pi_{21}/(\pi_{12} + \pi_{21}))$. Under $H_0 : \pi_{12} = \pi_{21}$ this is $\text{bin}(n^*, \frac{1}{2})$, and the normal approximation to it gives exactly McNemar's $z_0$.

**Proof.**

*Step 1.* A multinomial conditioned on the total of a subset of cells is multinomial on that subset with renormalized probabilities; for the two cells $(1, 2), (2, 1)$ this gives the binomial, with parameter $\frac{1}{2}$ under $H_0$.

*Step 2.* With mean $n^*/2$ and variance $n^*/4$, $z = (n_{21} - \frac{1}{2}n^*)/(\frac{1}{4}n^*)^{1/2} = (n_{21} - n_{12})/(n^*)^{1/2} = z_0$. ■

For Example 10.3, $P(\text{bin}(236, \frac{1}{2}) \geq 150) = 0.00002$ (one-sided).

### 10.1.5 Connection between McNemar and Cochran–Mantel–Haenszel tests

Each pair is its own $2 \times 2$ stratum (rows: occasion; columns: outcome; row totals $(1, 1)$): the $2 \times 2 \times n$ **subject-specific table**, versus the **population-averaged table**. Equal success probabilities at both occasions for every subject is conditional independence given subject, implying marginal homogeneity.

**Theorem 10.5 (McNemar as a CMH statistic)**

The Cochran–Mantel–Haenszel statistic for conditional independence in the $2 \times 2 \times n$ subject-specific table equals $(n_{21} - n_{12})^2/(n_{21} + n_{12})$.

**Proof.**

*Step 1.* Let $c_k \in \{0, 1, 2\}$ be the number of successes in stratum $k$. The hypergeometric moments are $E(n_{11k}) = c_k/2$ and $\text{Var}(n_{11k}) = c_k(2 - c_k)/4$.

*Step 2.* Concordant strata ($c_k = 0, 2$) have zero variance and $n_{11k} = E(n_{11k})$, so they contribute nothing.

*Step 3.* Discordant strata have variance $\frac{1}{4}$ and $n_{11k} - \frac{1}{2} = +\frac{1}{2}$ ($n_{12}$ strata) or $-\frac{1}{2}$ ($n_{21}$ strata), so

$$\text{CMH} = \frac{\left[\frac{1}{2}(n_{12} - n_{21})\right]^2}{\frac{1}{4}(n_{12} + n_{21})} = \frac{(n_{21} - n_{12})^2}{n_{21} + n_{12}}.$$

■

With more categories or occasions, generalized CMH tests with one stratum per subject test marginal homogeneity (Section 11.1.4).

## 10.2 Conditional logistic regression for binary matched pairs

**Motivation.** Conditioning away one parameter per subject yields $n_{21}/n_{12}$ for the subject-specific odds ratio.

### 10.2.1 Marginal versus conditional models for matched pairs

Let $(Y_1, Y_2)$ be a subject's pair (1 = success) and $x_1 = 0$, $x_2 = 1$.

**Definition (Marginal and conditional models)**

A **marginal** (population-averaged) model describes the margins, e.g. logit $P(Y_t = 1) = \alpha + \beta x_t$. A **conditional** (subject-specific) model has link $P(Y_{it} = 1) = \alpha_i + \beta x_t$, $i = 1, \ldots, n$.

In the marginal logit model $\hat{\beta} = \log[p_{+1}p_{2+}/(p_{+2}p_{1+})]$; in the conditional model $\beta$ is the association in the subject-stratified table.

**Proposition 10.6 (Subject-specific versus population-averaged effects)**

Let $\alpha_i$ be drawn from a distribution $F$ in the conditional model.
(a) With the identity link, $P(Y_2 = 1) - P(Y_1 = 1) = \beta$; the subject-specific and population-averaged effects coincide.
(b) With the logit link, the population-averaged probabilities are $P(Y_t = 1) = E_F[e^{\alpha+\beta x_t}/(1+e^{\alpha+\beta x_t})]$, which in general is not of the form $e^{a+bx_t}/(1 + e^{a+bx_t})$ with $b = \beta$.

**Proof.**

(a) Average $(\alpha_i + \beta) - \alpha_i = \beta$ over $F$.

(b) The first statement is total probability. Take $\alpha = \pm c$ with probability $\frac{1}{2}$ each and $\beta > 0$. Then $E[\text{expit}(\alpha)] = \frac{1}{2}$, so $b = \text{logit}\left(\frac{1}{2}[\text{expit}(c + \beta) + \text{expit}(\beta - c)]\right) \to \text{logit}(\frac{1}{2}) = 0$ as $c \to \infty$, while $\beta$ is fixed; so $b \neq \beta$ in general. ■

### 10.2.2 A logit model with subject-specific probabilities

The Cox–Rasch model is

$$\operatorname{logit} P(Y_{it}=1)=\alpha_i+\beta x_t, \qquad x_1=0,\ x_2=1, \tag{10.2}$$

so each subject's odds at occasion 2 are $e^{\beta}$ times those at occasion 1, responses independent given the parameters.

**Proposition 10.7 (Heterogeneity induces positive association)**

Suppose $\alpha_1,\dots,\alpha_n$ are i.i.d. from $F$ and, given $\alpha_i$, $Y_{i1},Y_{i2}$ are independent with the probabilities of (10.2). Then $\operatorname{Cov}(Y_{i1},Y_{i2})\geq 0$ for every $\beta$, with equality if and only if $F$ is degenerate.

**Proof.** [Supplement]

*Step 1.* With $f(a)=\operatorname{expit}(a)$, $g(a)=\operatorname{expit}(a+\beta)$, conditional independence gives $\operatorname{Cov}(Y_{i1},Y_{i2})=\operatorname{Cov}(f(\alpha),g(\alpha))$, $\alpha\sim F$.

*Step 2.* For an independent copy $\alpha'$, $f,g$ strictly increasing give $(f(\alpha)-f(\alpha'))(g(\alpha)-g(\alpha'))\geq 0$, whose expectation is $2\operatorname{Cov}(f(\alpha),g(\alpha))$.

*Step 3.* Equality forces $\alpha=\alpha'$ a.s., possible for independent copies only if $F$ is degenerate. ■

The model captures within-pair dependence, but its many $\alpha_i$ ruin ordinary ML (Proposition 10.9); use *conditional ML*.

### 10.2.3 Conditional ML inference for binary matched pairs

**Theorem 10.8 (Conditional ML estimator of the subject-specific odds ratio)**

In model (10.2) let $S_i=y_{i1}+y_{i2}$.
(a) $S_i$ is sufficient for $\alpha_i$. Given $S_i=0$ or $S_i=2$ the pair is determined; given $S_i=1$,

$$P(Y_{i1}=0,Y_{i2}=1\mid S_i=1)=\frac{e^{\beta}}{1+e^{\beta}}, \qquad P(Y_{i1}=1,Y_{i2}=0\mid S_i=1)=\frac{1}{1+e^{\beta}}.$$

(b) The conditional likelihood is $\ell_c(\beta)=(e^{\beta})^{n_{21}}/(1+e^{\beta})^{n^*}$ with $n^*=n_{12}+n_{21}$, i.e. $n_{21}\mid n^*\sim\operatorname{bin}(n^*,e^{\beta}/(1+e^{\beta}))$.
(c) The conditional ML estimator and its standard error are

$$\hat\beta=\log\frac{n_{21}}{n_{12}}, \qquad \mathrm{SE}=\Big(\frac{1}{n_{21}}+\frac{1}{n_{12}}\Big)^{1/2}.$$

**Proof.**

*Step 1.* The joint mass function is proportional to $\exp\big(\sum_i \alpha_i S_i+\beta\sum_i y_{i2}\big)$, an exponential family with $S_i$ sufficient for $\alpha_i$.

*Step 2.* Given $S_i\in\{0,2\}$ the pair is fixed. Given $S_i=1$, with $c_i=(1+e^{\alpha_i})(1+e^{\alpha_i+\beta})$, $P(0,1)=e^{\alpha_i+\beta}/c_i$ and $P(1,0)=e^{\alpha_i}/c_i$, so $P(0,1\mid S_i=1)=e^{\beta}/(1+e^{\beta})$.

*Step 3.* Multiplying over the $n^*$ discordant pairs gives $\ell_c(\beta)=e^{\beta n_{21}}/(1+e^{\beta})^{n^*}$, a binomial likelihood.

*Step 4.* $\log\ell_c$ is strictly concave with derivative $n_{21}-n^*e^{\beta}/(1+e^{\beta})$, zero at $e^{\hat\beta}=n_{21}/n_{12}$.

*Step 5.* The information $n^*e^{\beta}/(1+e^{\beta})^2$ equals $n_{21}n_{12}/n^*$ at $\hat\beta$; its inverse is $1/n_{21}+1/n_{12}$. ■

The conditional test of $\beta = 0$ is the binomial test of Proposition 10.4.

**Proposition 10.9 (Inconsistency of unconditional ML)**

Treating $\alpha_1, \ldots, \alpha_n$ as fixed parameters, the unconditional ML estimator of $\beta$ in (10.2) is $2\log(n_{21}/n_{12})$, which converges to $2\beta$.

**Proof.** [Supplement]

*Step 1.* A concordant pair is fitted perfectly by $\alpha_i \to \pm\infty$ (likelihood 1 for any $\beta$) and drops out.

*Step 2.* A $(1,0)$ pair has likelihood $p_1(1-p_2)$, $p_1 = \text{expit}(\alpha_i)$, $p_2 = \text{expit}(\alpha_i + \beta)$; the $\alpha_i$-score $(1-p_1) - p_2$ vanishes at $\alpha_i = -\beta/2$, giving $\text{expit}(-\beta/2)^2$. A $(0,1)$ pair profiles to $\text{expit}(\beta/2)^2$.

*Step 3.* The profile log-likelihood $2n_{21}\log\text{expit}(\beta/2) + 2n_{12}\log\text{expit}(-\beta/2)$ is that of Theorem 10.8 in $\beta/2$, so $\hat\beta/2 = \log(n_{21}/n_{12})$, which tends to $\beta$ by the law of large numbers. ■

### 10.2.4 Random effects in binary matched-pairs model

With $\alpha_i \sim N(\mu, \sigma^2)$ integrated out one gets a three-parameter GLMM (Chapter 12), which also gives $\hat\beta = \log(n_{21}/n_{12})$ when the sample log odds ratio is nonnegative.

**Proposition 10.10 (Mantel–Haenszel estimator)**

Applied to the $n$ subject-specific $2 \times 2$ tables, the Mantel–Haenszel estimator of the common odds ratio comparing observation 2 with observation 1 equals $n_{21}/n_{12}$.

**Proof.** [Supplement] With rows (observation 2, observation 1) and columns (success, failure), $n_{11k} = y_{k2}$, $n_{12k} = 1 - y_{k2}$, $n_{21k} = y_{k1}$, $n_{22k} = 1 - y_{k1}$, $n_{++k} = 2$, so

$$\hat\theta_{MH} = \frac{\sum_k n_{11k}n_{22k}/n_{++k}}{\sum_k n_{12k}n_{21k}/n_{++k}} = \frac{\sum_k y_{k2}(1-y_{k1})}{\sum_k (1-y_{k2})y_{k1}} = \frac{n_{21}}{n_{12}}.$$

■

| | $Y_2 = 1$ | $Y_2 = 0$ |
|---|---|---|
| $Y_1 = 1$ | $n_{11}$ | $n_{12}$ |
| $Y_1 = 0$ | $n_{21}$ | $n_{22}$ |

grey: $S_i \in \{0, 2\}$, no information on $\beta$

orange: $S_i = 1$, $n_{21} \mid n^* \sim \text{bin}\big(n^*, e^\beta/(1+e^\beta)\big)$

*Figure 10.1. Only the discordant cells carry conditional information about $\beta$.*

### 10.2.5 Logistic regression for matched case–control studies

In a matched case–control study each case $(Y = 1)$ is matched with a control $(Y = 0)$ and both are measured on $X$. With

$$\text{logit}\, P(Y_{it} = 1) = \alpha_i + \beta x_{it}, \tag{10.3}$$

$e^\beta$ is an $XY$ odds ratio and hence estimable from retrospective data.

**Example 10.11 (Diabetes and myocardial infarction)**

144 Navajo MI cases matched with 144 controls (age, gender). Diabetes: only case 37 pairs, only control 16, both 9, neither 82; $e^{\hat\beta} = 37/16 = 2.3$, $\hat\beta = 0.84$.

### 10.2.6 Conditional ML for matched pairs with multiple predictors

With $p$ predictors the model is

$$\text{logit}\, P(Y_{it}=1) = \alpha_i + \beta_1 x_{1it} + \cdots + \beta_p x_{pit} = \alpha_i + x_{it}'\boldsymbol{\beta}. \tag{10.4}$$

**Theorem 10.12 (Conditional likelihood as no-intercept logistic regression)**

In model (10.4), for pairs with $S_i = 1$,

$$P(Y_{i1}=0, Y_{i2}=1 \mid S_i=1) = \frac{e^{x_{i2}'\boldsymbol{\beta}}}{e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}}} = \frac{e^{x_i^{*\prime}\boldsymbol{\beta}}}{1+e^{x_i^{*\prime}\boldsymbol{\beta}}}, \qquad x_i^* = x_{i2} - x_{i1}.$$

Hence:
(a) the conditional ML estimate of $\boldsymbol{\beta}$ is the no-intercept logistic regression fit to the discordant pairs, with $y_i^* = 1$ for $(0,1)$, $y_i^* = 0$ for $(1,0)$ pairs and predictor $x_i^*$;
(b) a between-pair predictor ($x_{ji1} = x_{ji2}$ for all $i$) cancels from the conditional likelihood, so its coefficient is not estimable.

**Proof.**

*Step 1.* As in Theorem 10.8, the factors $e^{\alpha_i}$ and $c_i = (1+e^{\alpha_i + x_{i1}'\boldsymbol{\beta}})(1+e^{\alpha_i + x_{i2}'\boldsymbol{\beta}})$ cancel, leaving $e^{x_{i2}'\boldsymbol{\beta}}/(e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}})$; divide by $e^{x_{i1}'\boldsymbol{\beta}}$.

*Step 2.* The conditional likelihood $\prod_{S_i=1} \text{expit}(x_i^{*\prime}\boldsymbol{\beta})^{y_i^*}\left[1-\text{expit}(x_i^{*\prime}\boldsymbol{\beta})\right]^{1-y_i^*}$ is the no-intercept logistic likelihood, proving (a).

*Step 3.* If $x_{ji1} = x_{ji2}$ for all $i$, every $x_i^*$ has $j$th component 0 and $\beta_j$ disappears, proving (b). ■

In Example 10.11 (control $t=1$, case $t=2$) $y_i^* = 1$ and $x_i^* = -1$ (16 pairs) or $+1$ (37 pairs), again giving $\hat{\beta} = \log(37/16)$.

### 10.2.7 Marginal models and conditional models: extensions

Both kinds of model extend to multinomial responses and clusters; conditional ML estimates only within-cluster effects (Theorem 10.12(b)), random-effects models (Chapter 12) are not so restricted.

## 10.3 Marginal models for square contingency tables

**Motivation.** For $I > 2$, model the margins directly: cumulative logits (ordinal) or baseline-category logits (nominal).

For a square table with $\pi_{ab} = P(Y_1 = a, Y_2 = b)$, marginal homogeneity is $P(Y_1 = a) = P(Y_2 = a)$ for all $a$.

### 10.3.1 Marginal models for ordinal classifications

The cumulative logit model

$$\text{logit}\, P(Y_t \le j) = \alpha_j + \beta x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.5}$$

($x_1 = 0$, $x_2 = 1$) has proportional odds; $\beta > 0$ means $Y_1$ tends higher, $\beta = 0$ is marginal homogeneity. ML maximizes the multinomial likelihood in $\{\pi_{ab}\}$ under the margin constraints; df $= 2(I-1) - I = I - 2$.

### 10.3.2 Premarital and extramarital sex example

**Example 10.13 (Premarital and extramarital sex)**

1989 GSS: 475 subjects rated premarital and extramarital sex (1 = always wrong, ..., 4 = not wrong at all). Cumulative marginal proportions $(0.307, 0.389, 0.611)$ and $(0.815, 0.918, 0.987)$. $\hat{\beta} = 2.51$ (SE 0.13): far more permissive on premarital sex. Marginal homogeneity $G^2 = 348.1$ (df = 3), (10.5) $G^2 = 35.1$ (df = 2); better models in Section 10.4.7.

### 10.3.3 Marginal models for nominal classifications

The baseline-category model

$$\log \frac{P(Y_t = j)}{P(Y_t = I)} = \alpha_j + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.6}$$

is saturated for the margins; marginal homogeneity is $\beta_1 = \cdots = \beta_{I-1} = 0$, tested by $G^2$ or $X^2$ with df $= I - 1$. A non-iterative alternative uses $d_a = p_{+a} - p_{a+}$ and $\mathbf{d} = (d_1, \ldots, d_{I-1})'$.

**Proposition 10.14 (Bhapkar and Stuart statistics)**

(a) The covariance matrix $\mathbf{V}$ of $\sqrt{n}\,\mathbf{d}$ has elements

$$v_{ab} = -(\pi_{ab} + \pi_{ba}) - (\pi_{+a} - \pi_{a+})(\pi_{+b} - \pi_{b+}), \qquad a \neq b,$$
$$v_{aa} = \pi_{+a} + \pi_{a+} - 2\pi_{aa} - (\pi_{+a} - \pi_{a+})^2.$$

(b) Under marginal homogeneity the Wald (Bhapkar) statistic $W = n\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d}$ is asymptotically $\chi^2_{I-1}$. Stuart's statistic $W_0 = n\mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d}$ uses the null version $\hat{v}_{ab0} = -(p_{ab} + p_{ba})$, $\hat{v}_{aa0} = p_{+a} + p_{a+} - 2p_{aa}$, and

$$W = \frac{W_0}{1 - W_0/n}.$$

(c) For $I = 2$, $W_0$ is McNemar's statistic $z_0^2$.

**Proof.** [Supplement]

*Step 1.* $\mathbf{d}$ is the mean of i.i.d. $\mathbf{X}_k$ with $X_{ka} = \mathbb{1}\{Y_{k2} = a\} - \mathbb{1}\{Y_{k1} = a\}$. For $a \neq b$, $E(X_a X_b) = -(\pi_{ab} + \pi_{ba})$, and $E(X_a^2) = \pi_{+a} + \pi_{a+} - 2\pi_{aa}$; subtracting $E(X_a)E(X_b)$ gives (a).

*Step 2.* By the CLT and consistency of $\hat{\mathbf{V}}$, under $H_0$ ($E\mathbf{d} = \mathbf{0}$) $W \xrightarrow{d} \chi^2_{I-1}$.

*Step 3.* $\hat{\mathbf{V}} = \hat{\mathbf{V}}_0 - \mathbf{d}\mathbf{d}'$. By Sherman–Morrison with $q = \mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d} = W_0/n$,

$$\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} = q + \frac{q^2}{1-q} = \frac{q}{1-q},$$

so $W = W_0/(1 - W_0/n)$.

*Step 4.* For $I = 2$, $d_1 = p_{21} - p_{12}$ and $\hat{v}_{110} = p_{12} + p_{21}$, so $W_0 = (n_{21} - n_{12})^2/(n_{12} + n_{21})$. ■

For ordered categories a single-df ordinal test is often more powerful.

### 10.3.4 Migration example

**Example 10.15 (Migration 1980–1985)**

Region (NE, MW, S, W) in 1980 and 1985 for 55,981 U.S. residents; about 95% on the diagonal; e.g. 124 moved NE→W, 63 W→NE. Marginal homogeneity: $G^2 = 240.8$ (df = 3), Bhapkar $W = 236.5$. Margins $(0.218, 0.260, 0.330, 0.191)$ and $(0.213, 0.253, 0.339, 0.194)$ differ by $-0.005$, $-0.007$, $0.009$, $0.003$ (each $\pm 0.001$, Theorem 10.1): significant only because $n$ is huge.

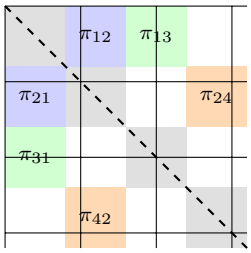
## 10.4 Symmetry, quasi-symmetry, and quasi-independence

**Motivation.** Quasi-symmetry fits well, has a subject-specific meaning and gives an LR test of marginal homogeneity.

**Definition (Symmetry)**

An $I \times I$ distribution satisfies **symmetry** if $\pi_{ab} = \pi_{ba}$ for all $a \neq b$.

Symmetry implies marginal homogeneity ($\pi_{a+} = \sum_b \pi_{ba} = \pi_{+a}$); for $I > 2$ the converse fails.




grey: diagonal, fitted exactly by S, QS, QI

mirror cells $(a, b), (b, a)$: equal under S; under QS, $\pi_{ab}/\pi_{ba} = e^{\tau_b - \tau_a}$

*Figure 10.2. Mirror pairs of a square table ($I = 4$).*

### 10.4.1 Symmetry as logit and loglinear models

For $\pi_{ab} > 0$, symmetry is $\log(\pi_{ab}/\pi_{ba}) = 0$ $(a < b)$, or, with $\mu_{ab} = n\pi_{ab}$,

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.7}$$

**Theorem 10.16 (Fitting and testing symmetry)**

For Poisson or multinomial counts:
(a) the ML fit of symmetry is $\hat{\mu}_{ab} = (n_{ab} + n_{ba})/2$ (so $\hat{\mu}_{aa} = n_{aa}$);
(b) the residual df is $I(I-1)/2$;
(c) the Pearson statistic is Bowker's statistic

$$X^2 = \sum_{a<b} \frac{(n_{ab} - n_{ba})^2}{n_{ab} + n_{ba}},$$

which for $I = 2$ is McNemar's $z_0^2$;
(d) the standardized Pearson residuals are $r_{ab} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2} = -r_{ba}$, and $\sum_{a<b} r_{ab}^2 = X^2$.

**Proof.**

*Step 1.* (a) With $\log \mu_{ab} = \lambda_{ab}$, $\lambda_{ab} = \lambda_{ba}$, the Poisson log-likelihood splits into $(n_{ab}+n_{ba}) \log \mu - 2\mu$ per pair $a < b$ and $n_{aa} \log \mu_{aa} - \mu_{aa}$ per diagonal cell; maximize each (same multinomial fit).

*Step 2.* (b) $\text{df} = I^2 - I - \binom{I}{2} = I(I-1)/2$.

*Step 3.* (c) With $m = (n_{ab}+n_{ba})/2$, both cells of a pair have $(n-m)^2 = (n_{ab}-n_{ba})^2/4$, contributing $2(n_{ab} - n_{ba})^2/(4m)$; diagonal cells contribute 0.

*Step 4.* (d) Each pair is fitted by one parameter with design $(1,1)'$ and equal weights, so the leverage is $h = \frac{1}{2}$ and $r_{ab} = (n_{ab} - m)/\sqrt{m/2} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2}$; squaring and summing gives (c). ■

### 10.4.2 Quasi-symmetry

**Definition (Quasi-symmetry)**

The **quasi-symmetry** (QS) model lets the main effects in (10.7) differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.8}$$

Symmetry is $\lambda_a^X = \lambda_a^Y$; independence is $\lambda_{ab} = 0$.

Since $\lambda_a^X + \lambda_b^Y = (\lambda_a^X + \lambda_b^X) + (\lambda_b^Y - \lambda_b^X)$, QS can be written

$$\log \mu_{ab} = \lambda'_{ab} + \tau_b, \qquad \lambda'_{ab} = \lambda'_{ba}, \quad \tau_b = \lambda_b^Y - \lambda_b^X, \quad \tau_I = 0. \tag{10.9}$$

**Proposition 10.17 (Likelihood equations for quasi-symmetry)**

The ML fit of QS satisfies

$$\hat{\mu}_{a+} = n_{a+}, \qquad \hat{\mu}_{+b} = n_{+b}, \qquad \hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba} \quad (a \le b);$$

one of the first two sets is implied by the other two. In particular $\hat{\mu}_{aa} = n_{aa}$, and the residual df is $(I-1)(I-2)/2$.

**Proof.**

*Step 1.* Poisson likelihood equations equate the sufficient statistics (row totals, column totals, symmetric sums $n_{ab} + n_{ba}$, $a \le b$) to their expectations; $a = b$ gives $\hat{\mu}_{aa} = n_{aa}$.

*Step 2.* Given the third set and the row equations, $\hat{\mu}_{+a} = \sum_b (n_{ab} + n_{ba} - \hat{\mu}_{ab}) = n_{a+} + n_{+a} - \hat{\mu}_{a+} = n_{+a}$.

*Step 3.* Form (10.9) has $I(I+1)/2 + (I-1)$ parameters, so $\text{df} = (I-1)(I-2)/2$. ■

QS has the multiplicative form

$$\pi_{ab} = \alpha_a \beta_b \gamma_{ab}, \qquad \gamma_{ab} = \gamma_{ba}, \tag{10.10}$$

with symmetry the case $\alpha_a = \beta_a$.

**Theorem 10.18 (Odds-ratio characterization of quasi-symmetry)**

Let all $\pi_{ab} > 0$.
(a) QS holds if and only if

$$\frac{\pi_{ab}\pi_{II}}{\pi_{aI}\pi_{Ib}} = \frac{\pi_{ba}\pi_{II}}{\pi_{bI}\pi_{Ia}} \qquad \text{for all } a, b. \tag{10.11}$$

(b) Under QS the local odds ratios satisfy $\theta_{ab} = \theta_{ba}$, where $\theta_{ab} = \pi_{ab}\pi_{a+1,b+1}/(\pi_{a,b+1}\pi_{a+1,b})$.

**Proof.** [Supplement]

*Step 1.* ($\Rightarrow$) Under (10.10) the $\alpha$'s and $\beta$'s cancel, leaving $\gamma_{ab}\gamma_{II}/(\gamma_{aI}\gamma_{Ib})$ and $\gamma_{ba}\gamma_{II}/(\gamma_{bI}\gamma_{Ia})$, equal by symmetry of $\gamma$.

*Step 2.* ($\Leftarrow$) Put $\alpha_a = \pi_{aI}$, $\beta_b = \pi_{Ib}/\pi_{II}$, $\gamma_{ab} = \pi_{ab}\pi_{II}/(\pi_{aI}\pi_{Ib})$; then $\pi_{ab} = \alpha_a\beta_b\gamma_{ab}$ and (10.11) says $\gamma_{ab} = \gamma_{ba}$.

*Step 3.* (b) Similarly $\theta_{ab} = \gamma_{ab}\gamma_{a+1,b+1}/(\gamma_{a,b+1}\gamma_{a+1,b})$, which is unchanged when $a, b$ are swapped since $\gamma$ is symmetric. ■

The meaning of QS comes from the subject-specific version of (10.6):

$$\log \frac{P(Y_{it} = j)}{P(Y_{it} = I)} = \alpha_{ij} + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.12}$$

with $x_1 = 0$, $x_2 = 1$, $\alpha_{iI} = \beta_I = 0$, and responses independent given the subject.

**Theorem 10.19 (Subject-specific logit models imply quasi-symmetry)**

If (10.12) holds and subjects are sampled from any distribution $F$ of $(\alpha_{i1}, \ldots, \alpha_{i,I-1})$, then the population-averaged table satisfies QS with $\tau_j = \lambda_j^Y - \lambda_j^X = \beta_j$ (constraint $\tau_I = 0$).

**Proof.** [Supplement]

*Step 1.* With $D_1(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k}$, $D_2(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k+\beta_k}$, conditional independence gives $P(Y_1 = a, Y_2 = b \mid \boldsymbol{\alpha}) = e^{\alpha_a} e^{\alpha_b+\beta_b}/[D_1(\boldsymbol{\alpha})D_2(\boldsymbol{\alpha})]$.

*Step 2.* Averaging over $F$, $\pi_{ab} = e^{\beta_b}\gamma_{ab}$ with $\gamma_{ab} = E_F[e^{\alpha_a+\alpha_b}/(D_1D_2)] = \gamma_{ba}$. This is (10.10) with $\alpha_a \equiv 1$, i.e. (10.9) with $\tau_b = \beta_b$. ■

Moreover (Conaway 1989) the conditional ML estimates of $\{\beta_j\}$ equal $\hat\lambda_j^Y - \hat\lambda_j^X$ from the QS fit.

### 10.4.3 Quasi-independence

**Definition (Quasi-independence)**

A square table satisfies **quasi-independence** (QI) if the classifications are independent given that the row and column outcomes differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \delta_a\, I(a = b). \tag{10.13}$$

The likelihood equations $\hat\mu_{a+} = n_{a+}$, $\hat\mu_{+a} = n_{+a}$, $\hat\mu_{aa} = n_{aa}$ fit the diagonal exactly; off-diagonal rectangle odds ratios equal 1, and df $= (I-1)^2 - I$ $(I \geq 3)$.

**Proposition 10.20 (Quasi-independence within quasi-symmetry)**

QI is QS with all $\gamma_{ab}$, $a \neq b$, in (10.10) equal; for $I = 3$ they coincide.

**Proof.** [Supplement]

*Step 1.* QI is (10.10) with $\gamma_{ab} = 1$ off the diagonal and $\gamma_{aa} = e^{\delta_a}$; conversely a constant off-diagonal $\gamma$ is absorbed into $\alpha$, and $\gamma_{aa}$ into $\delta_a$.

*Step 2.* For $I = 3$, the positive numbers $c_1 = (\gamma_{12}\gamma_{13}/\gamma_{23})^{1/2}$, $c_2 = (\gamma_{12}\gamma_{23}/\gamma_{13})^{1/2}$, $c_3 = (\gamma_{13}\gamma_{23}/\gamma_{12})^{1/2}$ satisfy $c_ac_b = \gamma_{ab}$ $(a \neq b)$. Then $\pi_{ab} = (\alpha_ac_a)(\beta_bc_b)$ off the diagonal, which is QI; with Step 1 the models coincide. ■

### 10.4.4 Migration revisited

**Example 10.21 (Migration: loglinear models)**

Example 10.15: independence ($G^2 = 125{,}923$) and symmetry ($G^2 = 243.6$, df $= 6$) fail; QI ($G^2 = 69.5$, df $= 5$) misfits NE→S and W→S; QS fits ($G^2 = 3.0$, df $= 3$), $\hat{\tau} = (-0.672, -0.623, 0.122)$ (W baseline): odds of living in the South rather than the West were $e^{0.122} = 1.13$ times higher in 1985.

### 10.4.5 Marginal homogeneity and quasi-symmetry

**Theorem 10.22 (Caussinus)**

For a table with all $\pi_{ab} > 0$,

$$\text{quasi-symmetry} + \text{marginal homogeneity} \iff \text{symmetry}.$$

**Proof.**

*Step 1.* ($\Leftarrow$) Symmetry is QS with $\alpha_a = \beta_a$ and implies marginal homogeneity.

*Step 2.* ($\Rightarrow$) Write $\pi_{ab} = \kappa_a \gamma^*_{ab}$ with $\kappa_a = \alpha_a/\beta_a$, $\gamma^*_{ab} = \beta_a\beta_b\gamma_{ab} = \gamma^*_{ba} > 0$. Marginal homogeneity $\kappa_j \sum_b \gamma^*_{jb} = \sum_a \kappa_a \gamma^*_{aj}$ reads

$$\kappa_j = \sum_a \kappa_a \, w_{aj}, \qquad w_{aj} = \frac{\gamma^*_{aj}}{\sum_b \gamma^*_{bj}} > 0, \quad \sum_a w_{aj} = 1.$$

*Step 3.* For $j$ with $\kappa_j = \min_a \kappa_a$, a positive-weight average equal to the minimum forces all $\kappa_a = \kappa$. Then $\pi_{ab} = \kappa\gamma^*_{ab} = \pi_{ba}$. ■

**Corollary 10.23 (Conditional test of marginal homogeneity)**

Given that QS holds, marginal homogeneity is equivalent to symmetry, i.e. to $\lambda^X_a = \lambda^Y_a$ (all $\tau_a = 0$). It is tested by

$$G^2(S \mid QS) = G^2(S) - G^2(QS), \qquad \text{df} = \tfrac{I(I-1)}{2} - \tfrac{(I-1)(I-2)}{2} = I - 1.$$

**Proof.** The equivalence is Theorem 10.22. Symmetry is the submodel $\tau_1 = \cdots = \tau_{I-1} = 0$ of (10.9), so the deviance difference is the LR statistic (df from Theorem 10.16(b) and Proposition 10.17). ■

For the migration data $G^2(S \mid QS) = 240.6$ (df $= 3$), close to $G^2 = 240.8$ and $W = 236.5$ of Example 10.15.

### 10.4.6 Ordinal quasi-symmetry model

With scores $u_1 \le \cdots \le u_I$, the **ordinal quasi-symmetry** model is

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \beta u_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}, \tag{10.14}$$

QS with $\tau_b = \beta u_b$; symmetry is $\beta = 0$. Equivalently

$$\log \frac{\pi_{ab}}{\pi_{ba}} = \beta(u_b - u_a), \qquad a < b, \tag{10.15}$$

a no-intercept logit model for $n_{ab} \mid (n_{ab} + n_{ba})$ with predictor $u_b - u_a$.

**Proposition 10.24 (Ordinal quasi-symmetry and marginal homogeneity)**

Let $u_1 < \cdots < u_I$ and all $\pi_{ab} > 0$.
(a) The likelihood equations are $\sum_a u_a \hat{\mu}_{a+} = \sum_a u_a n_{a+}$, $\sum_b u_b \hat{\mu}_{+b} = \sum_b u_b n_{+b}$, $\hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba}$ $(a \le b)$: fitted row and column means, not margins, match.
(b) Under (10.14) the column mean exceeds the row mean iff $\beta > 0$ (smaller iff $\beta < 0$); so ordinal QS + marginal homogeneity = symmetry, and $\beta = 0$ tests marginal homogeneity (df = 1).

**Proof.** [Supplement]

*Step 1.* (a) The sufficient statistics of $\log \mu_{ab} = \lambda'_{ab} + \beta u_b$ are the symmetric sums and $\sum_b u_b n_{+b}$, giving the second and third sets. Summing $(u_a + u_b)$ times the third set over $a \le b$ gives $\sum_{a,b}(u_a + u_b)\hat{\mu}_{ab} = \sum_{a,b}(u_a + u_b)n_{ab}$; subtracting the second set gives the first.

*Step 2.* (b) By (10.15),

$$\sum_{a,b}(u_b - u_a)\pi_{ab} = \sum_{a<b}(u_b - u_a)\pi_{ba}(e^{\beta(u_b - u_a)} - 1),$$

every term having the sign of $\beta$. Equal means force $\beta = 0$, i.e. symmetry; the converse is immediate. ■

### 10.4.7 Premarital and extramarital sex revisited

**Example 10.25 (Premarital and extramarital sex: symmetry models)**

Example 10.13: symmetry fails ($G^2 = 402.2$, df = 6); QS ($G^2 = 1.4$, df = 3) and ordinal QS with scores $1, \ldots, 4$ ($G^2 = 2.1$, df = 5, $\hat{\beta} = -2.86$) fit. By (10.15), a premarital response $x$ categories more permissive than the extramarital one is $e^{2.86x}$ times as likely as the reverse.

### 10.4.8 Other ordinal models for square tables

The **conditional symmetry** model (McCullagh 1978)

$$\log(\pi_{ab}/\pi_{ba}) = \tau \qquad \text{for all } a < b \tag{10.16}$$

describes $\pi_{ab} > \pi_{ba}$ for all $a < b$ (or the reverse); symmetry is $\tau = 0$. The **quasi-uniform association** model

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \beta u_a u_b + \delta_a I(a = b) \tag{10.17}$$

adds linear-by-linear association off the diagonal to QI; it is a special case of QS. For Example 10.13, $\hat{\tau} = -4.130$ (SE 0.451) and $\hat{\beta} = 0.632$ (SE 0.106).

## 10.5 Measuring agreement between observers

**Motivation.** Agreement is not association; it needs diagonal-oriented models and indices such as kappa.

**Example 10.26 (Pathologists' diagnoses)**

Pathologists A, B classified 118 slides into four ordered categories (1 = negative, ..., 4 = invasive carcinoma). Diagonal $(22, 7, 36, 10)$; main disagreements: B rated 3 when A rated

2 (14) or 4 (17). Independence fits poorly ($G^2 = 118.0$, df $= 9$), large positive diagonal residuals.

### 10.5.1 Agreement: departures from independence

$\sum_a \pi_{aa}$ is the agreement probability. Strong agreement needs strong association, not conversely (A might always rate one category higher). Independence is chance agreement.

### 10.5.2 Using quasi-independence to analyze agreement

For two subjects rated $a$ and $b$ by one observer and $b, a$ or $a, b$ by the other, the odds of agreement are

$$\tau_{ab} = \frac{\pi_{aa}\pi_{bb}}{\pi_{ab}\pi_{ba}} = \frac{\mu_{aa}\mu_{bb}}{\mu_{ab}\mu_{ba}}. \tag{10.18}$$

**Proposition 10.27 (Agreement odds under QI and QS)**

Under QI, $\tau_{ab} = \exp(\delta_a + \delta_b)$. Under QS, $\tau_{ab} = \exp(\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba})$.

**Proof.** [Supplement] In $\log \mu_{aa} + \log \mu_{bb} - \log \mu_{ab} - \log \mu_{ba}$ the main effects $\lambda_a^X + \lambda_a^Y + \lambda_b^X + \lambda_b^Y$ cancel against those of the off-diagonal cells, leaving $\delta_a + \delta_b$ under (10.13) and $\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba}$ under (10.8). ■

For Example 10.26, QI has $G^2 = 13.2$ (df $= 5$) and $\hat{\tau}_{23} = e^{2.5} = 12.3$.

### 10.5.3 Quasi-symmetry and agreement modeling

QS also allows association given disagreement and fits Example 10.26 well ($G^2 = 1.0$, df $= 2$, $\hat{\tau}_{23} = 10.7$); remove pairs with $n_{ab} + n_{ba} = 0$ (fitted 0) for correct df. Symmetry fits poorly ($G^2 = 39.2$, df $= 5$); $G^2(S \mid QS) = 38.2$ (df $= 3$): strong marginal heterogeneity.

### 10.5.4 Kappa measure of agreement

**Definition (Cohen's kappa)**

With $P_o = \sum_a \pi_{aa}$ and $P_e = \sum_a \pi_{a+}\pi_{+a}$ (agreement expected under independence),

$$\kappa = \frac{\sum_a \pi_{aa} - \sum_a \pi_{a+}\pi_{+a}}{1 - \sum_a \pi_{a+}\pi_{+a}} = \frac{P_o - P_e}{1 - P_e}.$$

$\kappa = 0$ at chance agreement, $\kappa = 1$ at perfect agreement.

**Theorem 10.28 (Asymptotic variance of sample kappa)**

Under multinomial sampling, $\sqrt{n}(\hat{\kappa} - \kappa) \xrightarrow{d} N(0, \sigma_\kappa^2)$, where

$$\sigma_\kappa^2 = \frac{P_o(1-P_o)}{(1-P_e)^2} + \frac{2(1-P_o)[2P_oP_e - \sum_a \pi_{aa}(\pi_{a+} + \pi_{+a})]}{(1-P_e)^3} + \frac{(1-P_o)^2[\sum_a \sum_b \pi_{ab}(\pi_{b+} + \pi_{+a})^2 - 4P_e^2]}{(1-P_e)^4}.$$

The estimated variance of $\hat{\kappa}$ (Fleiss, Cohen and Everitt 1969) substitutes sample proportions

and divides by $n$.

**Proof.** [Supplement]

*Step 1.* By the delta method for the multinomial, $\sigma^2 = \sum_{a,b}\pi_{ab}g_{ab}^2 - \left(\sum_{a,b}\pi_{ab}g_{ab}\right)^2$ with $g_{ab} = \partial\kappa/\partial\pi_{ab}$.

*Step 2.* With $\kappa = 1-(1-P_o)/(1-P_e)$, $\partial P_o/\partial\pi_{ab} = I(a=b)$ and $\partial P_e/\partial\pi_{ab} = \pi_{+a}+\pi_{b+}$,

$$g_{ab} = \frac{I(a=b)(1-P_e) - (1-P_o)(\pi_{+a}+\pi_{b+})}{(1-P_e)^2}.$$

*Step 3.* Since $\sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+}) = 2P_e$, $\sum\pi_{ab}g_{ab} = [P_o(1-P_e) - 2P_e(1-P_o)]/(1-P_e)^2$. With $s = \sum_a \pi_{aa}(\pi_{a+}+\pi_{+a})$ and $t = \sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+})^2$,

$$(1-P_e)^4\sum_{a,b}\pi_{ab}g_{ab}^2 = P_o(1-P_e)^2 - 2(1-P_e)(1-P_o)s + (1-P_o)^2 t.$$

*Step 4.* Subtracting the square of the first moment,

$$(1-P_e)^4\sigma_\kappa^2 = P_o(1-P_o)(1-P_e)^2 + 2(1-P_e)(1-P_o)(2P_oP_e - s) + (1-P_o)^2(t-4P_e^2),$$

which is the stated formula. ■

For Example 10.26, $P_o = 0.636$, $P_e = 0.281$, $\hat\kappa = 0.493$ (SE 0.057).

### 10.5.5 Weighted kappa: quantifying disagreement

**Definition (Weighted kappa)**

With weights $0 \le w_{ab} \le 1$, $w_{aa} = 1$, $w_{ab} = w_{ba}$,

$$\kappa_w = \frac{\sum_a\sum_b w_{ab}\pi_{ab} - \sum_a\sum_b w_{ab}\pi_{a+}\pi_{+b}}{1 - \sum_a\sum_b w_{ab}\pi_{a+}\pi_{+b}}.$$

Common choices are $w_{ab} = 1 - |a-b|/(I-1)$ and (Fleiss–Cohen) $w_{ab} = 1-(a-b)^2/(I-1)^2$.

Both kappas depend strongly on the margins; models of the agreement structure say more.

### 10.5.6 Extensions to multiple observers

With $R$ observers one can model the $\binom{R}{2}$ pairwise marginal tables simultaneously, use generalized kappas, or use latent-class models (Chapter 13).